

DEVA ANAND

devaanand@umass.edu • (413) 315-1715 • devaanand.com • linkedin.com/in/devaaa • github.com/Deva-1903

Education

MS, Computer Science, University of Massachusetts Amherst 2025–2027
Advanced NLP, Advanced Machine Learning, Optimization in CS, Systems for Data Science, Performance Engineering

BE, Computer Science and Engineering, Vels Institute of Science, Technology and Advanced Studies 2019–2023

Professional Experience

Wysa, Backend Engineer | *Bengaluru, India* Jul 2023 – Jul 2025

- Engineered a multi-tenant NHS eTriage backend (Node.js, MongoDB) for **8+ UK clients**, integrating the external Mayden iaptus platform via REST and processing **10,000+ monthly submissions**; contributed to **\$340K+** in revenue.
- Built a full-stack internal tool with a simple UI for the AI team to access and label data (React, Node.js, MongoDB), cutting manual work by **100+ hours/month** and lifting throughput by **60%**.
- Improved backend reliability by monitoring production logs, investigating incidents to root cause, and adding integration tests, auth hardening, and rate limiting; remediated **20+ VAPT findings**.
- Served as primary technical POC, mentored a second developer, and piloted a COO-sponsored GitHub Copilot experiment to automate tech-debt and migration work.

Cario Growth Services, Backend Developer Intern | *Chennai, India* Nov 2022 – May 2023

- Designed and shipped **20+ RESTful APIs** (Node.js, Fastify, PostgreSQL) with CRUD operations, input validation, and clear contracts used in production by the frontend team.
- Ramped quickly in an existing codebase, migrating core repositories from JavaScript to TypeScript, and shipped a production LLM feature (**4-bit quantized Llama-7B via llama.cpp**) with async queued generation.

Freelance / Open Source, Full-Stack Engineer | *Remote* Jul 2022 – Aug 2023

- Built and deployed **Underdogs Fitness** (Website), a production MERN platform with cron-automated workflows and git-webhook auto-deploy, independently end to end; live **3+ years**.

Projects

Spark ETL Performance Optimization (GitHub) PySpark, Spark UI, SQL, NYC TLC Dataset

- Built and optimized a Spark **data ingestion** pipeline over a **~1.4B-row** dataset with a 6-rule **data-quality** filter, broadcast joins, skew-join handling, partitioning, and column pruning, cutting end-to-end runtime **~25%** and the tail-latency ratio from **2.14x to 1.74x**, profiling stage metrics via Spark UI.

AgenticSearch – Provenance-First Entity Discovery (GitHub | Demo) Python, FastAPI, OpenAI / Groq, SQLite

- Built a Python/FastAPI service exposing **API endpoints** for an entity-discovery pipeline that turns queries into structured tables via typed retrieval planning and deterministic extraction, with provider routing, pipeline instrumentation, and a reviewer-facing data-access UI; hardened with **150+ tests**.

cf-ai-research-scout – Edge-Native AI Assistant (GitHub) TypeScript, Cloudflare Workers, Durable Objects, D1

- Built a service in **TypeScript** with durable, restart-safe workflows and D1 storage that keep multi-step jobs **reliable**, surviving process restarts and individual tool failures, and streaming results over WebSocket on Cloudflare Workers.

Publications

- “Alzheimer’s Disease Classification using Transfer Learning,” IEEE CONIT 2023, first author, deep transfer learning on neuroimaging data. (Paper)

Skills

Languages & Databases: Python, JavaScript, TypeScript, Java, SQL, PostgreSQL, MongoDB, SQLite

Backend & APIs: FastAPI, Node.js, Express, REST APIs, CRUD, HTTP, React, Next.js

Data & Pipelines: PySpark, Spark, ETL, data ingestion, data quality, batch pipelines

Ops & Tooling: monitoring, logging, instrumentation, automation, Docker, Git, GitHub Actions, CI/CD

Cloud & Platforms: AWS, GCP, Azure, Linux, MongoDB Atlas

Achievements

- Selected for **GirlScript Summer of Code 2023**; contributed to Linkfree, Freehit, and ProjectsHut. - Tutored **30+ underprivileged students** in programming at Sayur, a non-profit in Tamil Nadu.